

Week - 05 - Classes, Graphics\Lecture_10_code_2.py

```
1  '''
2  This code demonstrates associations between Canon and Projectile classes via
3  aggregation mechanism.
4  Class Canon receives a Projectile object (bomb) in its constructor which is created out of
5  Canon class. When Canon class stores this received object in its own instance variable
6  (self.bomb), it creates an aggregation association.
7  '''
8
9  import math
10 import time
11
12 G = 9.82
13 AIR_DRAG = 0.4
14
15 class Projectile:
16     def __init__(self, ID, size, energy, x_pos=0, y_pos=0):
17         self.id = ID
18         self.size = size
19         self.energy = energy
20         self.x_pos = x_pos
21         self.y_pos = y_pos
22
23     def get_projectile_data(self):
24         # A getter function
25         return (f'id {self.id}, size {self.size}, energy {self.energy} and position
26         ({self.x_pos}, {self.y_pos})')
27
28 class Canon:
29     def __init__(self, cid, length, type, bomb):
30         self.cid = cid
31         self.length = length
32         self.type = type
33         self.bomb = bomb
34
35     def fire(self, velocity, theta):
36         x_vel = velocity * math.cos(theta)
37         y_vel = velocity * math.sin(theta)
38         dt = 0.02
39
40         while self.bomb.y_pos >= 0:
41             self.bomb.x_pos += x_vel * dt
42             self.bomb.y_pos += y_vel * dt
43             x_vel -= AIR_DRAG * dt
44             y_vel -= G * dt
45             time.sleep(0.2)
46
47             animate(self.bomb.x_pos, self.bomb.y_pos)
48
49 # Helper function to display trajectory of our projectile
50 def animate(x, y):
51     width, height = 60, 30
52     scale = 100
```

```
50     screen = [" " for _ in range(width)] for _ in range(height)
51
52     column = round(x * scale)
53     row = height - 1 - round(y * scale)
54     if 0 <= column < width and 0 <= row < height:
55         screen[row][column] = "*"
56
57     print("\033[2J\033[H", end="")
58     print("\n".join("".join(line) for line in screen), flush=True)
59     time.sleep(0.1)
60
61 bomb_1 = Projectile('B101', 20, 15)
62
63 gun_1 = Canon('Gun-13', 20, 'Ground-based muzzle', bomb_1)
64 gun_1.fire(2, math.radians(45))
```